

SCIENTIFIC INVESTIGATIONS

Combining a wireless radar sleep monitoring device with deep machine learning techniques to assess obstructive sleep apnea severity

Shang-Yang Lin, PhD¹; Cheng-Yu Tsai, PhD^{2,3}; Arnab Majumdar, PhD²; Yu-Hsuan Ho, PhD⁴; Yu-Wen Huang, MS⁴; Chun-Kai Kao, MS⁵; Shang-Min Yeh, MS⁴; Wen-Hua Hsu, MS¹; Yi-Chun Kuan, MD^{6,7,8,9}; Kang-Yun Lee, MD, PhD³; Po-Hao Feng, MD, PhD³; Chien-Hua Tseng, MD, PhD³; Kuan-Yuan Chen, MD³; Jiunn-Horng Kang, MD, PhD^{10,11,12}; Hsin-Chien Lee, MD¹³; Cheng-Jung Wu, MD, PhD¹⁴; Wen-Te Liu, MD, PhD^{1,3,6,10}

¹School of Respiratory Therapy, College of Medicine, Taipei Medical University, Taipei, Taiwan; ²Department of Civil and Environmental Engineering, Imperial College London, London, United Kingdom; ³Division of Pulmonary Medicine, Department of Internal Medicine, Taipei Medical University-Shuang Ho Hospital, New Taipei City, Taiwan;

⁴Advanced Technology Lab, Wistron Corporation, Taipei, Taiwan; ⁵Wireless Technology and Antenna Research and Development Department, Wistron Corporation, Taipei, Taiwan; ⁶Sleep Center, Taipei Medical University-Shuang Ho Hospital, New Taipei City, Taiwan; ⁷Department of Neurology, Taipei Medical University-Shuang Ho Hospital, New Taipei City, Taiwan; ⁸Department of Neurology, School of Medicine, College of Medicine, Taipei Medical University, Taipei, Taiwan; ⁹Taipei Neuroscience Institute, Taipei Medical University, Taipei, Taiwan; ¹⁰Research Center of Artificial Intelligence in Medicine, Taipei Medical University, Taipei, Taiwan; ¹¹Department of Physical Medicine and Rehabilitation, Taipei Medical University Hospital, Taipei, Taiwan; ¹²Graduate Institute of Nanomedicine and Medical Engineering, College of Biomedical Engineering, Taipei Medical University, Taipei, Taiwan; ¹³Department of Psychiatry, Taipei Medical University Hospital, Taipei, Taiwan; ¹⁴Department of Otolaryngology, Taipei Medical University-Shuang Ho Hospital, New Taipei City, Taiwan

Study Objectives: The gold standard for diagnosing obstructive sleep apnea (OSA) is polysomnography (PSG). However, PSG is a time-consuming method with clinical limitations. This study aimed to create a wireless radar framework to screen the likelihood of 2 levels of OSA severity (ie, moderate-to-severe and severe OSA) in accordance with clinical practice standards.

Methods: We conducted a prospective, simultaneous study using a wireless radar system and PSG in a Northern Taiwan sleep center, involving 196 patients. The wireless radar sleep monitor, incorporating hybrid models such as deep neural decision trees, estimated the respiratory disturbance index relative to the total sleep time established by PSG (RDI_{PSG_TST}), by analyzing continuous-wave signals indicative of breathing patterns. Analyses were performed to examine the correlation and agreement between the RDI_{PSG_TST} and apnea-hypopnea index, results obtained through PSG. Cut-off thresholds for RDI_{PSG_TST} were determined using Youden's index, and multiclass classification was performed, after which the results were compared.

Results: A strong correlation ($\rho = 0.91$) and agreement (average difference of 0.59 events/h) between apnea-hypopnea index and RDI_{PSG_TST} were identified. In terms of the agreement between the 2 devices, the average difference between PSG-based apnea-hypopnea index and radar-based RDI_{PSG_TST} was 0.59 events/h, and 187 out of 196 cases (95.41%) fell within the 95% confidence interval of differences. A moderate-to-severe OSA model achieved an accuracy of 90.3% (cut-off threshold for RDI_{PSG_TST} : 19.2 events/h). A severe OSA model achieved an accuracy of 92.4% (cut-off threshold for RDI_{PSG_TST} : 28.86 events/h). The mean accuracy of multiclass classification performance using these cut-off thresholds was 83.7%.

Conclusions: The wireless-radar-based sleep monitoring device, with cut-off thresholds, can provide rapid OSA screening with acceptable accuracy and also alleviate the burden on PSG capacity. However, to independently apply this framework, the function of determining the radar-based total sleep time requires further optimizations and verification in future work.

Keywords: apnea-hypopnea index (AHI), respiratory disturbance index relative to the total sleep time established by PSG (RDI_{PSG_TST}), obstructive sleep apnea (OSA), polysomnography (PSG), wireless radar sleep monitoring device

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BRIEF SUMMARY

Current Knowledge/Study Rationale: The gold standard for diagnosing obstructive sleep apnea is polysomnography. However, polysomnography is a time-consuming method with clinical limitations.

Study Impact: This study introduces a wireless radar framework for screening the risk of having obstructive sleep apnea. This novel approach offers rapid, accurate obstructive sleep apnea screening, potentially reducing the clinical burden of polysomnography.

INTRODUCTION

Obstructive sleep apnea (OSA) is a sleep disorder characterized by partial or full airway collapse that results in restricted or interrupted airflow during sleep.¹ OSA has become a public health concern worldwide, affecting approximately 1 billion

people aged between 30 and 65 years, of which 425 million have moderate-to-severe OSA.² Additionally, OSA is linked to several comorbidities, such as an increased risk of developing cognitive impairments, cardiovascular disease, and metabolic disorders.^{3,4} Early diagnosis and timely intervention are crucial for managing OSA effectively.

In-laboratory polysomnography (PSG) is the gold standard for diagnosing OSA; it measures various physiological parameters during sleep, including brain activity, eye movement, muscle tone, heart rate, and breathing patterns.⁵ Clinically, PSG can be performed to detect respiratory events (eg, apnea and hypopnea), so that the apnea-hypopnea index (AHI) can be calculated by dividing the cumulative number of respiratory events by the total sleep time (TST), thereby enabling the classification of OSA severity.⁶ However, the limitations of PSG in a clinical setting may affect the reliability of PSG. That is, the time-consuming and labor-intensive nature of PSG may limit its effectiveness. In contrast to the short examination time required for radiology imaging studies, PSG requires prolonged monitoring and trained technologists, resulting in a long waiting time (estimated to be 2–60 months) for patients undergoing sleep studies.⁷ Because of the shortcomings of PSG, several alternative methods for screening the risk of OSA have been proposed. For example, a home sleep test, which involves the use of a portable monitoring device to record the heart rate, breathing patterns, oxygen levels, and snoring behavior of a patient, has been proposed as a more accessible and less expensive alternative to PSG for assessing OSA manifestations.⁸ However, this method may not be accurate for patients with complicated underlying health conditions or those who are easily awakened by the attached sensors or cables used in the home sleep test.⁹ To mitigate the interference caused by hardware devices, adapters, and cables during PSG for sleep studies, wireless techniques should be developed.

Researchers have proposed using sensing arrays containing triaxial accelerometers and pressure sensors to detect apnea events.¹⁰ A study developed image-based models by employing infrared cameras and motion sensor networks to observe the occurrence of sleep-disordered breathing.¹¹ However, these wireless models are still in the initial stages of development and are subject to several limitations, including factors that may affect the accuracy of results, such as the quality of vision and the effects of small movements (eg, eye blinking). Relatedly, wireless radar methods for detecting vital signs, such as Doppler or ultra-wideband radar, have been extensively explored in the literature, and these methods have potential applications in sleep monitoring.^{12,13} For example, in a study involving 176 participants, researchers developed a novel wireless system based on ultra-wideband radar for sleep apnea detection.¹⁴ Another study suggested the use of impulse-radio ultra-wideband radar as a wireless system to assess the severity of OSA by using the sleep data obtained from 94 participants.¹⁵ However, these proposed methods primarily utilized ultra-wideband frequency, which is costly and requires substantial hardware for transmitting and receiving radar signals.¹⁶ Specifically, a comparison of the characteristics and cost of various radar-based methods suggested that continuous-wave radar is more suitable than other methods for practical application because of its cost-effectiveness and adequate accuracy in detecting human physiological motion. A study recruited 11 participants and recorded their vital signs by using PSG and continuous-wave radar,¹⁷ and the measured respiration rates obtained using the 2 methods were reported to be highly correlated ($\rho = 0.91$). In a study with 44 participants, neural networks

were developed by combining continuous-wave radar signals with a convolutional recurrent strategy to obtain PSG results.¹⁸ Their results revealed strong correlations between the PSG-based AHI and estimated AHI results obtained using various proposed models (all $\rho > 0.81$). These observations suggest that radar-based sleep monitoring can be a contactless and low-cost method for screening OSA instead of methods that require commercial PSG equipment; however, the results of that study were based on a small participant pool, which did not exhibit an adequate distribution of OSA severity. Thus, further investigations with more participants and a more diverse distribution of OSA severity are required to verify the wireless radar framework for sleep monitoring and clarify its feasibility.

The present prospective study aimed to develop an OSA severity screening framework by combining wireless radar sleep monitoring devices (ie, a continuous-wave radar module) with deep learning methods. The hypothesis of the present study is that recorded radar signals represent the respiratory patterns associated with sleep-disordered breathing, and that analyzing such signals through trained deep learning models can aid in the evaluation of OSA severity. In the present study, deep learning models were trained using radar continuous-wave signals and PSG parameters from labeled respiratory event data. A total of 196 participants were recruited from a sleep center in Northern Taiwan, and the correlations between the sleep-disordered index results obtained through PSG and a proposed wireless radar framework were determined. Additionally, the present study investigated the classification performance of the proposed framework for screening the likelihood of 3 levels of OSA severity (ie, mild-to-severe, moderate-to-severe, and severe OSA) on the basis of several cut-off thresholds. The goal of this study was to develop a novel surrogate method for PSG that provides less interference while producing adequately accurate and robust results. The results of the present study can elucidate the feasibility of using a wireless radar-based sleep monitoring device instead of PSG.

METHODS

Study protocol

The protocol for the present prospective study was examined and approved by the Institutional Review Board of the Office of Human Research at Taipei Medical University (TMU-SHH: N202105113). The specifications and safety certifications of the employed device were provided to the ethics committee for review. The participants of the present study were fully informed of the research details and asked to consent in writing prior to data collection. All data processing, analysis, and maintenance procedures were conducted in accordance with the approved and established protocols.

Participant recruitment

The present prospective study was conducted at a sleep center in Northern Taiwan (Taipei Medical University-Shuang Ho Hospital, New Taipei City, Taiwan) between July 2021 and January 2022. Patients who reported snoring or were suspected

of having OSA were referred to the sleep center for PSG and were invited to participate in the present study. The exclusion criteria were as follows: (1) being younger than 20 or older than 80 years, (2) regular use of psychotropic or hypnotic medications in the past half year, (3) having a history of otorhinolaryngological surgery for sleep disorder treatment, (4) being pregnant, and (5) having pulmonary diseases that may require the use of oxygen during a PSG examination (eg, chronic obstructive pulmonary disease). As presented in **Figure 1**, the wireless radar device was installed in PSG rooms so that wireless radar sleep monitoring could be implemented following the PSG protocol without the participants wearing additional devices or performing other specific procedures. The measurements obtained through PSG and wireless radar monitoring were automatically synchronized on the basis of the start time of the PSG for further comparisons.

PSG parameters and data scoring

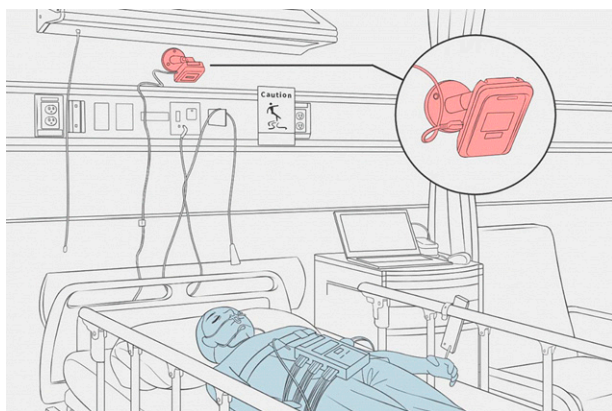
In the present prospective study, the recording of in-laboratory PSG outcomes was performed using two systems, namely the Embla MPR (ResMed Global Supplier Alliance, Sydney, Australia) and ResMed Embla N7000 (ResMed, San Diego, California). The PSG outcomes were displayed and scored using RemLogic software (version 3.41; Embla Systems, Thornton, Colorado); this process was performed by registered polysomnographic technologists in accordance with the 2017 PSG guidelines and the updated standard diagnostic criteria of the American Academy of Sleep Medicine.¹⁹ To mitigate scoring bias and to ensure the accuracy of sleep staging and respiratory events, all PSG outcomes were independently reviewed by at least 2 certified technologists who underwent regular inter-scoring training. For inconsistent scoring outcomes, further discussion was conducted to reach a final consensus regarding the scoring results. For respiratory events, obstructive apnea was defined as the cessation of airflow for at least 10 seconds despite ongoing respiratory effort, which can occur because of airway occlusion, causing the peak thermal sensor excursion to

decrease by more than 90% from the baseline; by contrast, central apnea was defined as the cessation of airflow with the absence of respiratory effort. Hypopnea was characterized as a greater than 30% reduction in nasal pressure for at least 10 seconds accompanied by arousal or oxygen desaturation of more than 3%. AHI was calculated by dividing the cumulative number of apnea and hypopnea events by the TST, and OSA severity was classified as mild (AHI = 5 to 15 events/h), moderate (AHI = 15 to 30 events/h), or severe (AHI $\geq$ 30 events/h).²⁰

Wireless radar framework

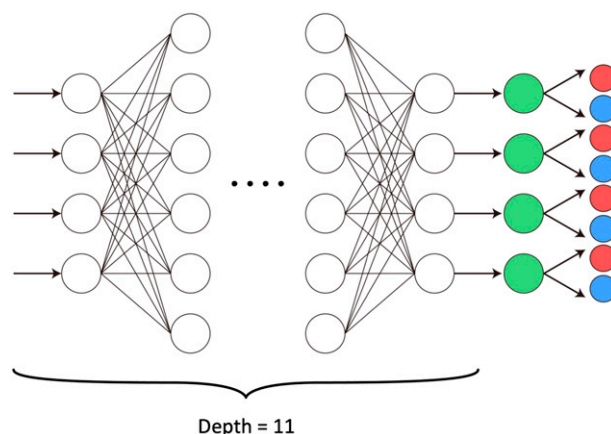
The present study used wireless radar sleep monitoring devices developed by Wistron Medical Technology Corporation (New Taipei City, Taiwan). The radar device has received safety approval in the motion detection wireless category from the Taiwan National Communications Commission (CCAF19LP2510T5) and medical device approval in the noncontact monitoring system category from the Taiwan Ministry of Health and Welfare (no. 007955). The device incorporates radio frequency modules to capture continuous-wave signals. These signals are processed by hybrid models, specifically deep neural decision trees that combine machine learning and deep learning features. Technically, the device utilizes a 24-GHz continuous wave with a wavelength of approximately 12.5 mm to monitor chest wall movements. Chest displacement due to respiration and cardiac activity typically range from 1–12 mm and from 0.1–0.5 mm, respectively. When the patient is positioned 1–1.5 m away from the device, the 24-GHz radar is the optimal frequency band for measuring physiological signals clinically. **Figure 2** depicts the architecture of the hybrid models designed to predict respiratory events and sleep stage based on radar signals and derived features (eg, respiratory rate variability).²¹ First, for the respiratory event prediction, the hyperparameters for these binary classification models were set as follows: a learning rate of 0.05, a batch size of 400, 200 training epochs, 1 decision tree, a neural network depth of 11, and a used features rate of 0.5.^{22,23} After these models were optimized, they were trained

Figure 1—Data collection.



Sleep parameters were collected through polysomnography, and a wireless radar framework was employed to capture continuous-wave signals indicating breathing patterns.

Figure 2—Hybrid model architecture based on radar signals for respiratory event prediction.



The developed hybrid model consisted of a neural network with a depth of 11, followed by a binary decision tree.

and validated on a data set comprising 29,496 instances (duration of 30 seconds). This dataset comprised both apnea and hypopnea events ($n = 15,264$) as well as normal respiratory data without any respiratory events ($n = 14,232$), obtained from approximately 100 individuals (male:female ratio, 3:2) in various sleep positions. **Table S1** in the supplemental material describes the overall performance of the established models, which collectively achieved an accuracy rate of approximately 75.78%, a specificity of 76.19%, and a sensitivity of 75.34%. **Figure 3** illustrates the signal waveforms of various respiratory events obtained through the PSG and wireless radar framework. The radar devices emit radio frequency signals targeting a participant lying on a bed and capture the corresponding signals that are reflected back during PSG recording. These signals are subjected to entropy calculations to extract distinct patterns, such as peak-to-peak intervals, valley-to-valley intervals, and wave peak magnitude, providing the approximate respiratory patterns. The resulting features are fed into a hybrid model to predict the total number of abnormal breathing patterns (eg, respiratory events). Subsequently, the models aggregate the predicted number of events and divide this by the TST as determined by PSG to calculate a radar-based index, known as the respiratory disturbance index (RDI_{PSG_TST}), which is then used in following analyses. For detailed specifications, installation guidelines, and usage precautions related to the device (eg, caution regarding signal errors), please consult the Appendix in the supplemental material.

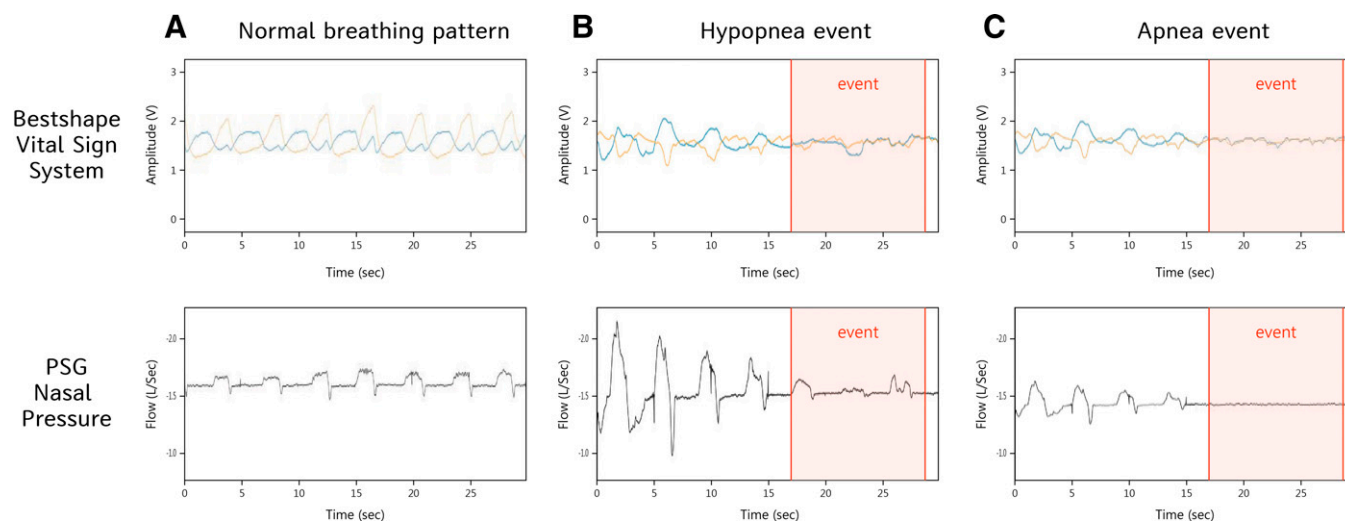
The proposed wireless framework enables the estimation of in-bed periods and discerns sleep stages, specifically wake, rapid eye movement, and non-rapid eye movement, enabling an independent assessment of TST. The establishment of the sleep stage classification models (ie, hybrid models) followed a procedure similar to that used in the training of the respiratory event models (neural network depth: 13). These models were trained and validated on a dataset that included wake ($n = 9,833$), non-rapid eye movement

($n = 11,503$), and rapid eye movement ($n = 8,160$) stages from the same individuals as the respiratory event data. **Figure S1** presents a comparison example of sleep stage classification between PSG and the radar system. Notably, although the proposed framework enables the estimation of TST and thereby calculates the respiratory disturbance index based on that ($RDI_{Radar_Based_TST}$), the accuracy of this function requires further refinement and empirical validation in future studies. Consequently, the current research primarily focused on comparing the AHI and the RDI_{PSG_TST} , ensuring alignment with the corresponding time intervals as determined by PSG.

Statistical analysis

The open-source statistics library Scikit-learn (version 0.21.2), which is implemented in Python, was used to analyze data and generate descriptive statistical summaries.²⁴ The descriptive statistics for baseline characteristics, PSG parameters, and RDI results are expressed as means and standard deviations (SDs). Pearson correlation analysis was conducted to determine the relationships between AHI (PSG-based) and RDI, and the statistical significance level was set at $P < .05$. Additionally, the present study used a Bland–Altman plot, which is a graphical method for comparing 2 quantitative measurements and evaluating the agreement in terms of the accuracy between PSG and wireless radar device results.²⁵ For generating the Bland–Altman plot, the difference between data points and the average of each data point were determined and presented against x and y axes. Furthermore, the Bland–Altman plot featured a horizontal line that indicated the mean difference and 2 additional lines that indicated the upper and lower limits of agreement, which were calculated as the mean difference ± 1.96 times the SD of the differences. The differences within these limits were used to identify the systematic bias between 2 devices and to assess the degree of agreement.²⁶ Subsequently,

Figure 3—Signal waveforms of various respiratory events as obtained through polysomnography and wireless radar framework (Bestshape vital sign system).



Signal waveforms corresponding to normal breathing patterns (A), hypopnea events (B), and apnea events (C).

on the basis of clinical practice considerations, the present study compared severity classification performance at 2 levels. According to clinical practice guidelines, active intervention is recommended for patients with moderate-to-severe OSA (AHI ≥ 15 events/h),²⁷ whereas curative intervention is strongly recommended for those with severe OSA (AHI ≥ 30 events/h).²⁸ Therefore, the present study explored the ability of the wireless radar device to classify and differentiate between moderate-to-severe OSA (AHI ≥ 15 events/h or AHI < 15 events/h) and severe OSA (AHI ≥ 30 events/h or AHI < 30 events/h).²⁹ Specifically, the present study first compared the classification performance of the 2 devices by applying PSG cut-off thresholds, namely the AHI thresholds of ≥ 15 and ≥ 30 events/h. Subsequently, a receiver operating characteristic curve analysis was conducted to determine the optimal cut-off thresholds for the RDI results with the application of Youden's index.³⁰ These established cut-off thresholds were used to determine whether a patient had moderate-to-severe OSA (AHI ≥ 15 events/h or AHI < 15 events/h) or severe OSA (AHI ≥ 30 events/h or AHI < 30 events/h). Furthermore, the present study determined and compared the multiclassification performance (ie, normal-to-mild, moderate, and severe OSA) achieved using 2 sets of cut-off thresholds (AHI and RDI_{PSG_TST}).

RESULTS

Baseline characteristics and sleep parameters of participants

The present study enrolled 196 patients, whose baseline characteristics, PSG-based sleep parameters, and RDI results are presented in **Table 1**. Most of the participants were male, and their mean age was 44.79 years (SD = 13.46 years). Regarding the sleep architecture metrics determined by PSG, the mean sleep efficiency was 76.86% (SD = 11.29%), and the TST was 284.26 minutes (SD = 41.32 minutes). The results for the OSA-associated indices are as follows: the mean AHI was 31.07 events/h (SD = 25.55 events/h) and the oxygen desaturation level was 23.02 events/h (SD = 26.12 events/h). Among the enrolled participants, 22 (11.2%) had no OSA, 45 (23%) had mild OSA, 49 (25%) had moderate OSA, and 80 (40.8%) had severe OSA. For the radar-based index, the mean radar-based TST was 292 minutes (SD = 33.83 minutes). The mean RDI obtained through radar device measurement and divided by the TST determined by PSG (RDI_{PSG_TST}) was 30.45 events/h (SD = 19.29 events/h). The mean RDI obtained through radar device measurement and divided by the TST determined by the radar system (RDI_{Radar-Based_TST}) was 30.71 events/h (SD = 18.93 events/h).

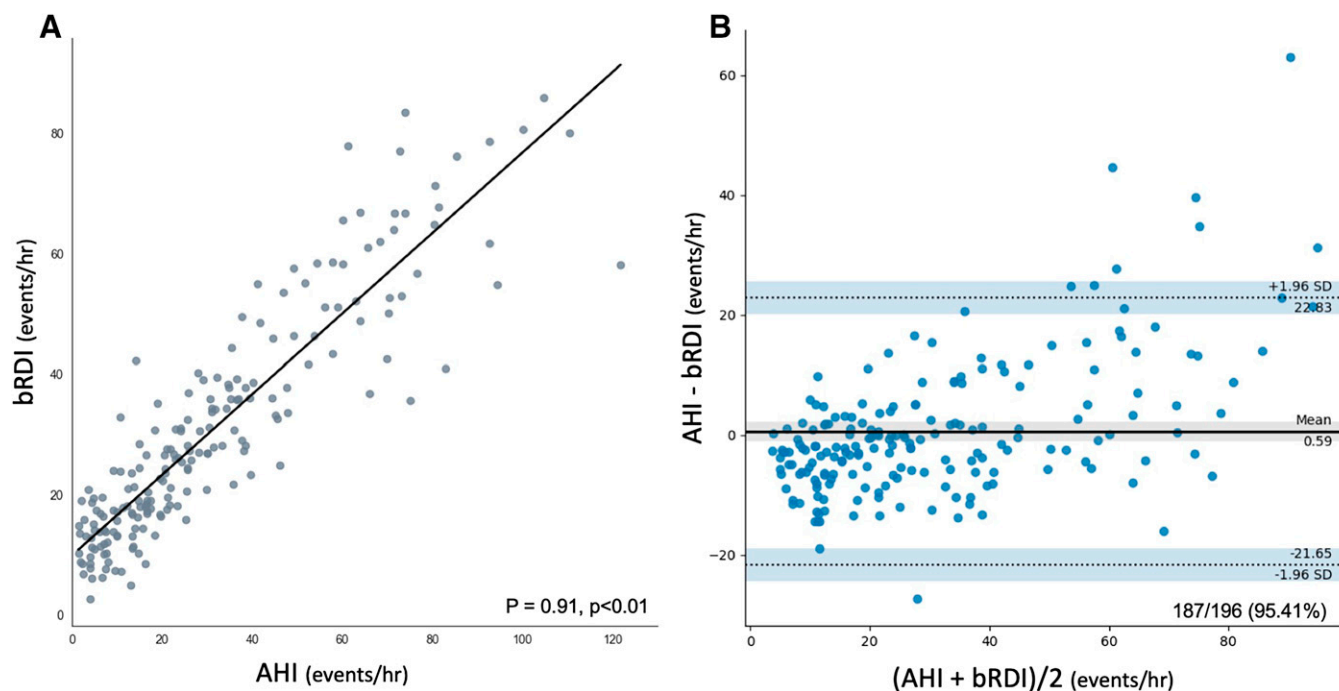
Correlation and agreement

Figure 4 presents the correlation and agreement between the PSG-based AHI and radar-based RDI_{PSG_TST} results. As presented in **Figure 4A**, the 2 measures were strongly correlated ($\rho = 0.91$,

Table 1—Baseline characteristics and sleep parameters of participants (n = 196).

Characteristics and Sleep Parameters	
Baseline characteristics	
Age (years)	44.79 $\pm$ 13.46
Sex (male/female)	146/50
BMI (kg/m ²)	29.44 $\pm$ 42.92
Neck circumference (cm)	37.52 $\pm$ 4.46
Waist circumference (cm)	92.28 $\pm$ 12.83
Sleep stage summary	
Sleep efficiency (%)	76.86 $\pm$ 11.29
Mean SpO ₂ (%)	95.52 $\pm$ 2.45
Minimum SpO ₂ (%)	84.32 $\pm$ 10.55
WASO (minutes)	55.74 $\pm$ 35.64
TST (minutes)	284.26 $\pm$ 41.32
Sleep stage (% of SPT)	
Wake	16.29 $\pm$ 10.0
REM	70.93 $\pm$ 9.39
NREM	13.14 $\pm$ 7.58
Snoring	
Snoring event (events)	841.7 $\pm$ 1022.4
Snoring index (events/h)	181.08 $\pm$ 216.19
Sleep quality index (events/h)	
AHI	31.04 $\pm$ 25.55
AI	7.48 $\pm$ 15.27
HI	23.56 $\pm$ 18.84
ODI	23.02 $\pm$ 26.12
Arl	27.37 $\pm$ 14.86
OSA severity, n (%)	
Normal	22 (11.2%)
Mild	45 (23%)
Moderate	49 (25%)
Severe	80 (40.8%)
Radar-based index (events/h)	
Radar-based TST	292.59 $\pm$ 33.83
RDI _{PSG_TST}	30.45 $\pm$ 19.29
RDI _{Radar-Based_TST}	30.71 $\pm$ 18.93

Data are expressed as mean $\pm$ standard deviation unless otherwise indicated. AHI = apnea-hypopnea index, AI = apnea index, ArI = arousal index, BMI = body mass index, HI = hypopnea index, NREM = non-rapid eye movement, ODI = oxygen desaturation index for $\geq 3\%$, OSA = obstructive sleep apnea, RDI_{PSG_TST} = respiratory disturbance index obtained through radar device measurement and divided by the total sleep time determined by polysomnography, RDI_{Radar-Based_TST} = respiratory disturbance index obtained through radar device measurement and divided by the total sleep time determined by the radar system, REM = rapid eye movement, SpO₂ = peripheral capillary oxygen saturation, SPT = total sleep period time, TST = total sleep time, WASO = wake time after sleep onset.

Figure 4—Correlation and Bland–Altman plots for measured sleep index results.

(A) Correlation between apnea-hypopnea index results derived from polysomnography (PSG) and respiratory disturbance index (RDI_{PSG_TST}) results derived from wireless radar sleep monitoring device (Bestshape vital sign system) and divided by the TST determined by PSG. **(B)** Degree of agreement between results from polysomnography and wireless radar framework.

$P < .01$). The mean difference between the PSG-based AHI and radar-based RDI_{PSG_TST} results was 0.59 events/h (SD = 11.35 events/h; **Figure 4B**). The 95% confidence interval for the differences was between -21.65 and 22.83 events/h, and the results of 95.4% (187/196) of the participants were within this range.

Classification performance of wireless radar based on different cut-off thresholds

Table 2 presents the classification performance results of the 3 severity models that were obtained using PSG-based AHI

cut-off thresholds. Model A, which was used to detect mild-to-severe OSA ($AHI \geq 5$ events/h), achieved an accuracy of 89.8% (SD = 4.24%) and an area under the curve (AUC) of 54.55% (SD = 6.97%). Model B, which was used to detect moderate-to-severe OSA ($AHI \geq 15$ events/h), achieved an accuracy of 83.2% (SD = 5.2%) and an AUC of 76.8% (SD = 5.9%). Model C, which was used to detect severe OSA ($AHI \geq 30$), achieved an accuracy of 91.8% (SD = 3.8%) and an AUC of 91.4% (SD = 3.9%). **Table 3** presents the classification performance of the models that were obtained using

Table 2—Results from wireless radar sleep monitoring device (Bestshape vital sign system) for screening and classifying risk of moderate-to-severe and severe OSA using polysomnography severity criteria.

Variables	Model A (Mild-to-Severe OSA)	Model B (Moderate-to-Severe OSA)	Model C (Severe OSA)
Number	AHI ≥ 5 (n = 174) AHI < 5 (n = 22)	AHI ≥ 15 (n = 129) AHI < 15 (n = 67)	AHI ≥ 30 (n = 80) AHI < 30 (n = 116)
Precision (%)	89.69 $\pm$ 4.26	81.2 $\pm$ 5.5	91.0 $\pm$ 4.0
Recall (%)	100.0 $\pm$ 0.0	96.9 $\pm$ 2.4	88.8 $\pm$ 4.4
Accuracy (%)	89.8 $\pm$ 4.24	83.2 $\pm$ 5.2	91.8 $\pm$ 3.8
F1 score (%)	94.57 $\pm$ 3.17	88.3 $\pm$ 4.5	89.9 $\pm$ 4.2
AUC (%)	54.55 $\pm$ 6.97	76.8 $\pm$ 5.9	91.4 $\pm$ 3.9

Data are expressed as mean $\pm$ standard deviation. AHI = apnea-hypopnea index, AUC = area under the curve, OSA = obstructive sleep apnea.

Table 3—Results from wireless radar sleep monitoring device (Bestshape vital sign system) for screening and classifying risk of moderate-to-severe and severe OSA using cut-off thresholds.

Variables	Model A (Mild-to-Severe OSA)	Model B (Moderate-to-Severe OSA)	Model C (Severe OSA)
Cut-off (events/h)*	> 15.84	> 19.2	> 28.86
Number	RDI _{PSG_TST} ≥ 15.84 (n = 147) RDI _{PSG_TST} < 15.84 (n = 49)	RDI _{PSG_TST} ≥ 19.2 (n = 125) RDI _{PSG_TST} < 19.2 (n = 71)	RDI _{PSG_TST} ≥ 28.86 (n = 83) RDI _{PSG_TST} < 28.86 (n = 113)
Precision (%)	95.92 ± 2.77	94.4 ± 3.2	90.1 ± 4.2
Recall (%)	81.03 ± 5.49	90.7 ± 4.1	91.3 ± 4.0
Accuracy (%)	80.1 ± 5.59	90.3 ± 4.1	92.4 ± 3.7
F1 score (%)	87.85 ± 4.57	92.5 ± 3.7	90.7 ± 4.1
AUC (%)	76.88 ± 5.9	90.1 ± 4.2	92.2 ± 3.8

Data are expressed as mean ± standard deviation. *According to Youden's index. AUC = area under the curve, OSA = obstructive sleep apnea, RDI_{PSG_TST} = respiratory disturbance index obtained through radar device measurement and divided by the total sleep time determined by polysomnography.

the cut-off thresholds based on Youden's index. First, for the mild-to-severe OSA model, the applied cut-off threshold was 15.84 events/h, and 147 patients were revealed to have RDI_{PSG_TST} values that exceeded this threshold. This model had an accuracy of 80.1% (SD = 5.59%) and an AUC of 76.88% (SD = 5.9%). For the moderate-to-severe OSA model, the applied cut-off threshold was 19.2 events/h, and 125 patients were discovered to have RDI_{PSG_TST} values that exceeded this threshold. This model had an accuracy of 90.3% (SD = 4.1%) and an AUC of 90.1% (SD = 4.2%). A cut-off threshold of 28.86 events/h was applied for the severe OSA model, and 82 patients were discovered to have RDI_{PSG_TST} values that exceeded this threshold. The application of this model with this cut-off threshold resulted in an accuracy of 92.4% (SD = 3.7%) and an AUC of 92.2% (SD = 3.8%). Overall, the classification performance achieved with the application of the wireless radar sleep monitoring device with the RDI_{PSG_TST} cut-off thresholds was more favorable than that achieved using the PSG-based AHI cut-off thresholds.

Comparison of multiclass classification performance

Table 4 presents the multiclass classification performance achieved on the basis of the PSG-based AHI cut-off thresholds (5, 15, and 30 events/h). Specifically, an overall accuracy of

68.37%, an average recall of 56.58%, and a precision of 74.34% were achieved. **Table 5** presents the classification performance achieved using the RDI cut-off thresholds (15.84, 19.2, and 28.86 events/h); this model achieved an overall accuracy of 72.96%, an average recall of 66.93%, and a precision of 54.55%, representing an improvement over the performance of the model presented in **Table 4**. Overall, the results presented in **Table 4** and **Table 5** indicate that the wireless radar sleep monitoring devices produced superior classification performance with RDI_{PSG_TST} cut-off thresholds than with PSG-based AHI cut-off thresholds.

DISCUSSION

To address the clinical limitations and limited accessibility of PSG, the present study aimed to develop and validate wireless sleep monitoring systems that use radar devices and deep learning techniques to screen for OSA and determine OSA severity. This developed framework provides RDI_{PSG_TST} results by employing deep neural decision tree models and determining breathing patterns on the basis of continuous-wave signals, thereby enabling the evaluation of OSA risk. The obtained outcomes indicated strong correlations between the AHI and

Table 4—Results from wireless radar sleep monitoring device (Bestshape vital sign system) that employs PSG severity criteria to screen and classify risk of normal, mild, moderate, or severe OSA.

PSG	Wireless Radar Sleep Monitoring Device				Recall (%)
	Normal	Mild	Moderate	Severe	
Normal	2	14	6	0	9.09%
Mild	0	22	22	1	48.89%
Moderate	0	4	39	6	79.59%
Severe	0	0	9	71	88.75%
Precision (%)	100%	55%	51.32%	91.03%	68.37%

OSA = obstructive sleep apnea, PSG = polysomnography.

Table 5—Results from wireless radar sleep monitoring device (Bestshape vital sign system) that employs cut-off thresholds to screen and classify risk of normal, mild, moderate, or severe OSA.

PSG	Wireless Radar Sleep Monitoring Device				Recall (%)
	Normal	Mild	Moderate	Severe	
Normal	28	5	1	0	82.35%
Mild	16	11	5	1	33.33%
Moderate	5	6	31	7	63.27%
Severe	0	0	7	73	88.75%
Precision (%)	57.14%	50%	70.45%	90.12%	72.96%

OSA = obstructive sleep apnea, PSG = polysomnography.

RDI_{PSG_TST} results; they also revealed high measurement agreement between the results obtained through PSG and the wireless radar sleep monitoring device. Next, when the derived cut-off points were applied (moderate-to-severe OSA, 19.2 events/h; severe OSA, 28.86 events/h), classification performance was greater than 90% for both accuracy and AUC, and overall accuracy of 83.7% was achieved for multiclass classification. The developed framework can be applied as an alternative to PSG; its main advantage is its noncontact monitoring feature, which can reduce the risk of interference due to attached sensors or examination equipment.

Researchers have compared various PSG hardware options to evaluate their prediction performance. A study used a patch-type device with a single-lead electrocardiogram and 3-axis accelerometer to screen for OSA risk; its primary outcomes were AUCs of 0.82 and 0.91 for the moderate-to-severe OSA model and the severe OSA model, respectively.³¹ However, this type of device is not recommended for patients with atrial fibrillation, implanted pacemakers, ventricular tachycardia, or pregnancy because of the challenges of analyzing cardiac signals in the presence of excessive arrhythmia.³² However, the system proposed in the present study was wireless and, thus, less susceptible to disruptions in sleep data caused by contact sensors.³³ Overall, wireless radar methods for detecting vital signs may be suitable for sleep monitoring.

To compare ultra-wideband and continuous-wave methods, a study combined an ultra-wideband radar system with fast Fourier transform processing technology to measure the breathing rates of 15 participants, of which only 6 were diagnosed with mild-to-moderate OSA and none were diagnosed with severe OSA.³⁴ Although the error rate reported by that study for the breathing rate ranged between 4.1 and 8.5%, indicating a high level of accuracy, the study used a small sample size and did not include participants with severe OSA. Additionally, that study tested a system that focused solely on chest wall motion during respiration, potentially limiting its ability to accurately assess the effects of sleep position and the total number of aberrant respiratory events in its final results. Furthermore, for radar systems, the trade-off between frequency and sensitivity must be considered. Although higher frequencies often provide greater resolution and sensitivity, allowing for the detection of subtle target movements and reflections because of their shorter wavelengths, their penetration capability is limited. That is,

these high-frequency signals can be easily obstructed and reflected, and their strength diminishes over distance, potentially limiting their detection capability at long ranges. Conversely, lower frequencies can penetrate objects and clothing more effectively but produce less favorable results in terms of resolution and sensitivity. The wavelength of the 24-GHz continuous-wave radar used in the present study was approximately 12.5 mm, which is suitable for detecting typical respiratory and heart rate-induced chest movements that range from 1–12 mm and from 0.1–0.5 mm, respectively.^{35,36} Collectively, when the costs and benefits of employing an ultra-wideband method are considered, the 24-GHz continuous-wave radar used in the present study may be the optimal choice for performing clinical physiological signal measurements, particularly when a patient is positioned approximately 1–1.5 m from the measurement device and in the front of the bed.

The present study has several notable findings. First, the statistically strong correlation between the AHI and RDI_{PSG_TST} results and the high agreement between the measurements obtained from PSG and the proposed wireless radar framework suggest the feasibility of using the proposed system as a surrogate for PSG. Second, the present study revealed that the application of the cut-off thresholds for moderate-to-severe OSA and severe OSA produced OSA severity classification results that exhibited satisfactory prediction accuracy and adequate AUC values. Furthermore, the proposed system only comprised an ensemble radar sensor box that was combined with deep learning methods for the automated estimation of reasonably accurate RDI_{PSG_TST} results. Because of their wireless nature, radar devices are suitable for individuals whose sleep behaviors are easily disrupted by attached sensors; thus, they provide more accurate and representative results of the baseline sleep status. Additionally, radar devices are straightforward to install at home, and a technician is not required for continuous monitoring and result scoring through these devices; thus, they are a suitable option for home-based sleep monitoring, multiday examinations, and for analyzing comprehensive data to accurately and sensitively screen for OSA severity.

The present study, nevertheless, has several notable limitations. First, only a small number of female participants were recruited, which is a common limitation in sleep disorder studies. Second, the proposed wireless radar framework provided

RDI_{PSG_TST} results primarily on the basis of the continuous-wave signals generated by responses to chest wall movements, which represent breathing patterns. In other words, the proposed framework estimated RDI_{PSG_TST} without the support of electroencephalograms, which may theoretically compromise its ability to evaluate the frequency of sleep arousals. Future work should consider developing an arousal detection model based on radar signals to enhance the clinical value. Regarding sleep architecture, the proposed wireless framework, combined with hybrid models, currently provides functionality for estimating in-bed periods and predicting sleep stages (ie, wake, rapid eye movement, and non-rapid eye movement), thus enabling the calculation of radar-based TST. Subsequently, this framework enables the calculation of the RDI_{Radar_Based_TST} based on that. However, the accuracy of these functions requires further refinement and empirical validation in future studies. Third, the present study excluded patients diagnosed with chronic comorbidities and pulmonary diseases to mitigate the effects of altered breathing patterns (due to these underlying illnesses) on the results. However, other factors, such as smoking habits, individual craniofacial characteristics, oral hygiene, and upper airway obstruction anatomy, can still affect breathing patterns (eg, buccal respiration), and these factors should be considered when applying the proposed system.^{37,38} Next, the employed hybrid models were developed using data that encompassed multiple sleep positions. Although these models consider the effects of sleep position, the sleeping positions of an individual can notably affect the frequency of respiration events.³⁹ Thus, future studies should investigate the nuanced differences between the radar framework and PSG across multiple sleep positions. Another limitation of the present study is the absence of oximetry data. Although the proposed system did not require oximetry data, such data can be used to further clarify oxygen desaturation conditions because oximetry signals are strongly associated with OSA manifestations; therefore, the inclusion of these data could enable a more comprehensive evaluation of OSA severity without incurring significant costs.^{40,41} Given this, future studies employing the proposed framework should consider integrating oximetry measurements with the wireless radar system to enhance diagnostic precision and provide a more detailed assessment of OSA severity by accounting for respiratory events alongside oxygen saturation levels. Next, the primary function of the current models was to identify respiratory events, specifically apnea and hypopnea. However, these models did not adequately differentiate between central and obstructive apnea or hypopnea. Additionally, the present study did not analyze respiratory frequency or interbreath intervals to identify the patterns or duration of respiratory events. Although the radar system was proficient in detecting chest signals, it was not configured to detect limb movements, making it unsuitable for periodic limb movement index calculations. The detection range of the radar system can be expanded to encompass the entire body, thereby providing a more comprehensive sleep disorder assessment. Another potential limitation is the data collection bias in the current study, arising from its single-center setting and reliance on the manufacturer. The current hospital examination schedule, which

resulted in the limited mean TST, may not have captured the full spectrum of sleep architecture and could have potentially introduced bias in interpreting the severity of OSA. To mitigate these potential biases and improve the generalizability of present findings, future studies should be multicenter and conducted independently. This approach can aid in validating the robustness of the proposed framework, minimizing potential biases, and ensuring the integrity of the proposed models across various settings. Overall, the proposed radar device should be further validated. Specifically, future studies should consider adding channels (eg, pulse oximetry and heart rate monitoring), verify the sleep staging function, differentiate between various types of sleep apnea, such as central and obstructive sleep apnea, develop additional functions (eg, arousal detection and limb movement determination), and conduct independent trials in multiple centers to enhance the robustness, generalizability, and accuracy of the proposed device.

CONCLUSIONS

Given the limited availability of PSG and the need to address its drawbacks, the present prospective study aimed to develop and validate a wireless sleep monitoring system that combined a radar device with deep learning methods. Sleep and breathing pattern data were collected from 196 patients at a sleep center through PSG and the wireless radar sleep monitoring device. The correlation between AHI and RDI_{PSG_TST} and the agreement between these 3 OSA severity assessment models were explored. Furthermore, cut-off thresholds were applied to the proposed wireless framework, and the classification performance of the PSG and radar device models for OSA severity (ie, mild-to-severe, moderate-to-severe, and severe OSA) was compared. First, the results of the present study indicated a strong correlation between the AHI and RDI_{PSG_TST} results ($\rho = 0.91$, $P < .05$) and 95.4% agreement between the 2 models. For the binary classification model, an accuracy of 80.1% and an AUC of 76.88% were obtained for moderate-to-severe OSA screening when the RDI_{PSG_TST} cut-off threshold of ≥ 15.84 events/h was applied. Next, an accuracy of 90.3% and an AUC of 90.1% were obtained for moderate-to-severe OSA screening when the RDI_{PSG_TST} cut-off threshold of ≥ 19.2 events/h was applied. Similarly, for the severe OSA model, accuracy of 92.4% and an AUC of 92.2% were obtained when the RDI_{PSG_TST} cut-off threshold was applied (RDI_{PSG_TST} ≥ 28.86 events/h). Furthermore, when these 3 cut-off thresholds were applied, the multiclass classification model (normal, mild, moderate, and severe OSA) was determined to have an accuracy of 72.96%. Collectively, the wireless radar sleep monitoring devices and hybrid models (eg, deep neural decision trees) proposed in the present study can be used as an alternative to PSG for implementing rapid OSA risk screening. Moreover, the proposed wireless framework is capable of estimating in-bed periods and discerning sleep stages, thus enabling an independent assessment of TST. However, the accuracy of this functionality necessitates further optimization and empirical validation in future research efforts.

ABBREVIATIONS

AHI, apnea-hypopnea index

AUC, area under the curve

OSA, obstructive sleep apnea

PSG, polysomnography

RDI_{PSG_TST} , respiratory disturbance index obtained through radar device measurement and divided by the total sleep time determined by polysomnography

$RDI_{Radar-Based_TST}$, respiratory disturbance index obtained through radar device measurement and divided by the total sleep time determined by the radar system

SD, standard deviation

TST, total sleep time

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Address correspondence to: Wen-Te Liu, MD, PhD, School of Respiratory Therapy, College of Medicine, Taipei Medical University, No. 250, Wuxing Street, Taipei 110301, Taiwan; Tel: +886-2-2736-1661; Fax: +886-2-2739-1143; Email: b7801077@tmu.edu.tw or lion5835@gmail.com

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